

The maximum length of a chess game
under the FIDE Laws is 17,697 plies

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August 2026

0. Origin

a known value, and no proof

Where the problem comes from



Ing Chess, developer of Augment Chess



The longest possible chess game, and bounds on the number of possible chess games

by François Labelle

Abstract

The longest possible chess game is 8848.5 moves long. The number of possible chess games is at least 10^{2041} according to a Monte Carlo simulation, and at most 10^{1040} according to a calculation.

Introduction

What is the length of the longest possible chess game? It used to be infinite because the 50-move rule and the draw by 3-fold repetition aren't automatic—they must be claimed by one of the players. Some people answered the question assuming that a draw must be claimed if it is available, presumably to get a more interesting answer. We no longer need to assume anything: in 2014, FIDE introduced two new rules, the 75-move rule and the draw by 5-fold repetition, which are automatic. When introduced, the 5-fold repetition rule applied to consecutive repetition so it could be avoided by the players, but the 75-move rule cannot be avoided and makes the longest possible chess game finite, and the number of possible chess games finite. (In 2017, the 5-fold repetition rule has been modified to no longer require consecutive repetition, giving a second independent reason why chess is finite.)

Length calculation

Article 9.6a from the [Laws of Chess of 2014](#) says that the game is drawn if "any consecutive series of 75 moves have been completed by each player without the movement of any pawn and without any capture. If the last move resulted in checkmate, that last move prevails!" So at a high level, the play will be to play 14.5 moves without the movement of any pawn and without any capture, then move a pawn or make a capture to avoid an immediate draw, and repeat. There are $16 \times 8 = 128$ pawn moves available, and 30 captures available, giving us 158 "delay plys" (half-moves that can be used to delay the draw). For the pawn to cross past each other, 8 of them will have to capture as they move, so we're down to 150 delay plys. Another concern is that by waiting exactly 14.5 moves before playing a delay ply, Black gets to do all the delay plys, which won't work because White has to play delay plys too. Each time we need to switch control we'll have to play 74.0 moves between 2 delay plys instead of 75.5 moves. It's not too hard to find a way to schedule all 150 delay plys by switching control 4 times, but it's possible to be clever and switch control only 3 times. At the end we have a choice between capturing to get down to 2 kings and draw that way, or playing a non-capturing move and draw by the 75-move rule. Sometimes the second choice is the only option because the first choice may require switching control which costs 0.5 moves. Taking all of that into account, I believe that the length of the longest possible chess game is $118 \times 75.0 - 3 \times 0.5 = 8848.5$ moves.

For comparison with other values on the internet, the equivalent calculation when the players are forced to claim a draw by the 50-move rule gives $118 \times 50.0 - 3 \times 0.5 = 5898.5$ moves. This is in agreement with a section of the [German Wikipedia article](#) on the 50-move rule. According to [this forum posting](#), Karl Fabel obtained this number as early as 1947. The 50-move rule is different from the 75-move rule because it is not automatic, but also because Article 9.3a says that the ply triggering the 50-move rule will be written on the player's scorebook, and that the player intends to play it, but it doesn't clearly state that the ply is played or that it is part of the game. To avoid using 0.5 moves due to this, for a 50-move rule record I suggest using a plan which ends with a capture on the last move (without switching control a 4th time).

A specific plan (Deedlit and Tatzelwurm)

Below is an explicit schedule for the 150 delay plys. It switches control 3 times using the idea discussed by Deedlit and Tatzelwurm in [this chess.com forum](#)

F. Labelle, *The longest possible chess game*, 2015

"I believe that the length of the longest possible chess game is 8848.5 moves."

Labelle 2015

0. Origin

I. Rules

II. Counting

III. Switches

IV. Search

V. Checking

VI. Meaning

I–III the upper bound — hand-checkable combinatorics

IV–V computation — the search that preceded it, and the checking

VI what the result means

$$L \leq 150 \cdot 118 - 3 = \mathbf{17,697}$$

I. Rules

in 2014, chess became finite

The automatic terminations — in force since 1 July 2014

0. Origin

I. Rules

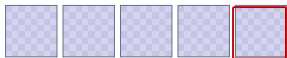
II. Counting

III. Switches

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VI. Meaning



×5

fivefold repetition

Art. 9.6.1

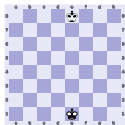
the same position, five times



75 moves by each player

Art. 9.6.2

the clock reaches 150



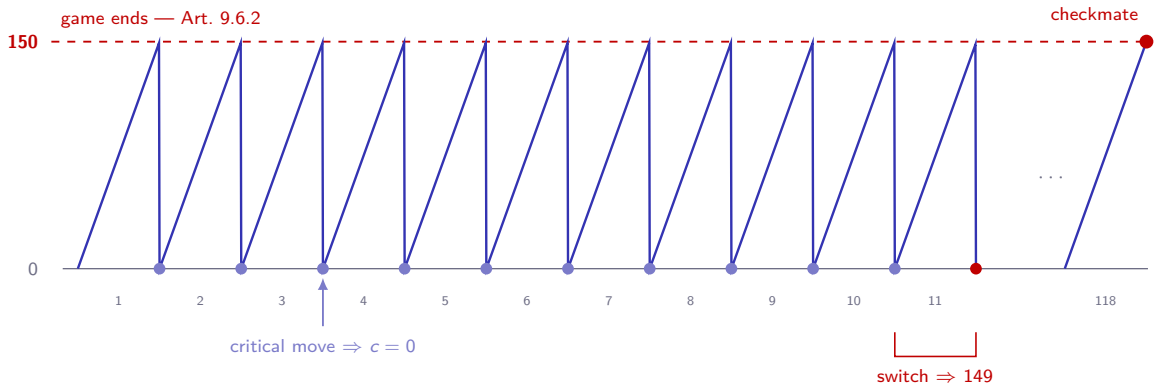
dead position

Art. 5.2.2

no continuation can mate

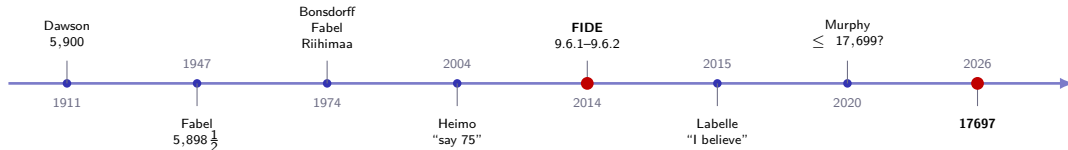
Before 2014 the threefold-repetition and 50-move draws had to be *claimed* — two cooperating players could play forever (Euwe 1929, via Thue–Morse).

The halfmove clock $c(t)$



$c(t)$ = plies since the last pawn move or capture · the reference game's first eleven segments, to scale

Prior work

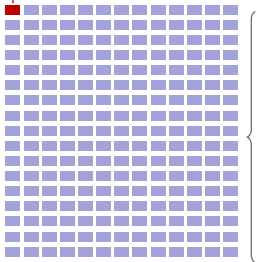


"We clearly need at least one switch ... but is it possible to do it with only two? If not, can this lower bound be proved?"

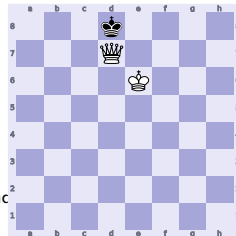
Murphy VII, SIGBOVIK 2020

The scale of 17,697 plies

a tournament game
 ≈ 80 plies



221 × tournament games, end to end
= **17697** plies the longest game



The final position — king and queen against king

The last ply is neither a pawn move nor a capture: a *quiet* checkmate,

with the clock at exactly 150

II. Counting

$K \leq 118$ — counting suffices

0. Origin

I. Rules

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VI. Meaning

Definition 2.1 (critical move)

A pawn move or a capture. (En passant is both; a promotion is a pawn move; castling is neither.) Any other move is *quiet*.

Definition 2.2 (segments · endpoints)

Let the critical moves fall at plies $t_1 < \dots < t_m$. Set $T = 1$ if quiet plies follow the last of them, else $T = 0$. The endpoints are $e_i = t_i$ and, if $T = 1$, $e_{m+1} = L$; and $K = m + T$.

Definition 2.3 (switches S)

The *colour* of e_i is the colour of whoever played that ply. The virtual endpoint $e_0 = 0$ is Black by convention.

$$S = \#\{i \leq K : \text{col}(e_i) \neq \text{col}(e_{i-1})\}.$$

Critical moves are exactly the moves that reset the clock, and they are *irreversible* — a pawn cannot retreat and a captured piece never returns.

Lemma 2.4 — the segment decomposition

Lemma 2.4 (segment decomposition)

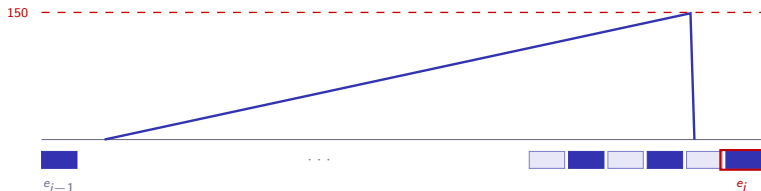
$$L = 150K - S - \sum_{i=1}^K \delta_i, \quad \delta_i \geq 0$$

δ_i is segment i 's shortfall from its target length (150 if its endpoints share a colour, 149 otherwise). In particular $L \leq 150K - S$.

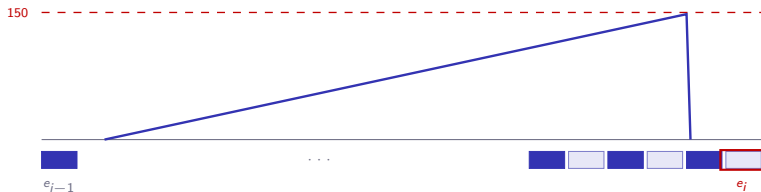
Proof

- (1) All plies of segment i except possibly the last are quiet. If its length were 151 or more, the first 150 would all be quiet and the game would already have ended there (Art. 9.6.2) — contradiction. So the length is ≤ 150 .
- (2) Plies alternate colour, so a segment's length is even exactly when its two endpoints share a colour. If they differ the length is odd, and being ≤ 150 it is ≤ 149 .
- (3) Summing the segment lengths telescopes to $e_K - e_0 = L$. □

Step (2) of Lemma 2.4 — parity



same colour
 $\Rightarrow$ even length $\Rightarrow$ 150

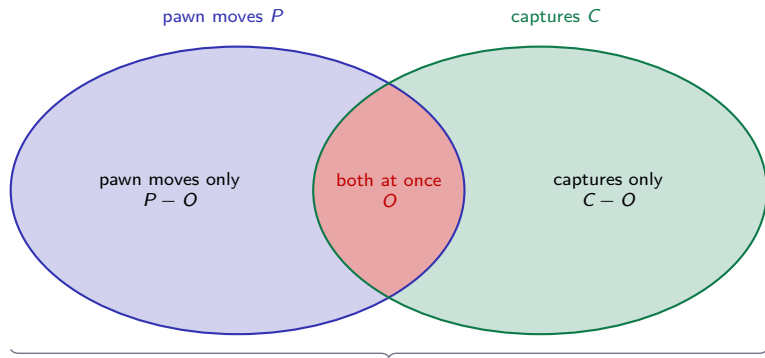


colour changes
 $\Rightarrow$ odd length $\Rightarrow$ 149

Plies alternate colour $\Rightarrow$ the parity of a segment's length is exactly whether its two endpoints share a colour

Counting the critical moves

$$K = (P - O) + (C + T)$$



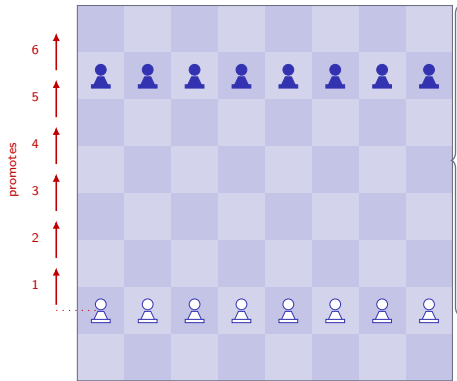
+ T
closing segment

$$K = (P - O) + (C + T)$$

$$\leq 88 + 30 = \mathbf{118}$$

P pawn moves C captures O both at once T closing segment $\in \{0, 1\}$

Lemma 3.1 — one pawn's budget



Lemma 3.1 (pawn basics)

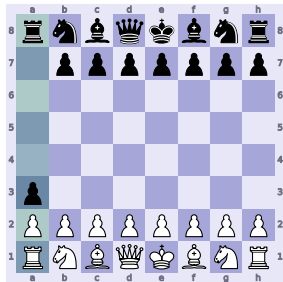
Each pawn makes at most 6 pawn moves, so $P \leq 96$. A pawn that makes exactly 6 advances exactly one rank each time and *promotes*.

Proof

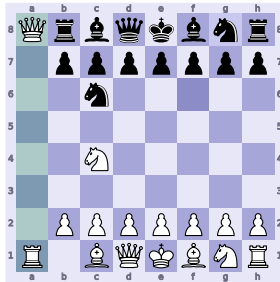
Every pawn move advances at least one rank, and there are six ranks between a pawn's home rank and the promotion rank. Six moves therefore means one rank each and arrival on the promotion rank, where promoting is compulsory. $\square$

Corollary: $P = 96$ is equivalent to “all 16 pawns walked up one rank at a time, with no double push, and all 16 promoted”.

Lemma 3.3 — unresolved origin pairs



after 4...a3: four Black



after a8=Q: six White

Definition 3.2 · Lemma 3.3

The two pawns starting on file i are its *origin pair*. If either ever makes a diagonal pawn move the file is *resolved*, else *unresolved*; write f for the number of resolved files.

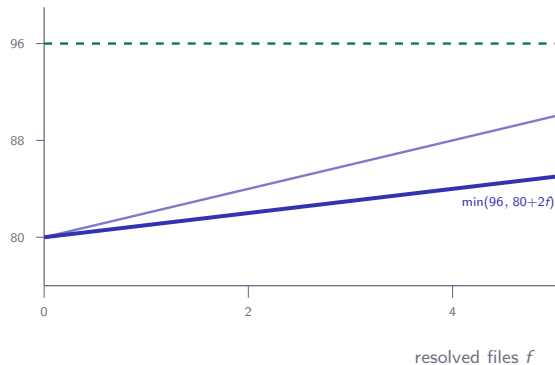
An unresolved pair makes at most 10 pawn moves combined, and 10 is attained.

Proof

While both are on the board they stand on ranks $2 + a$ and $7 - b$ and cannot pass, so $2 + a < 7 - b$, i.e. $a + b \leq 4$. Once one is captured (by some third piece) its contribution is frozen at $d \leq 4$ and the survivor has at most 6 by Lemma 3.1: $d + 6 \leq 10$. $\square$

Remark 3.4. “Four combined” holds only *while both are on the board*. Used as a whole-game bound it excludes legal games.

Proposition 3.6 — $P - O \leq 88$



Lemma 3.5 (overlap floor)

$O \geq f$. (For each resolved file pick one diagonal pawn move by a pawn of that origin file; distinct files have distinct movers, hence distinct overlaps.)

Proposition 3.6

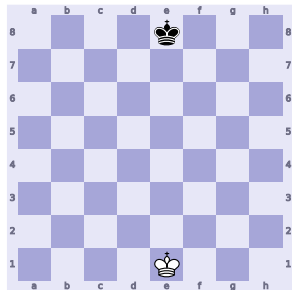
Since $P \leq \min(96, 80 + 2f)$,

$$P - O \leq \min(96, 80 + 2f) - f \leq \mathbf{88},$$

with equality only at $f = 8$, $P = 96$, $O = 8$.

Maximising over f is what makes this a statement about *every game* rather than about games that happen to have eight overlaps.

Lemma 3.7 — $C + T \leq 30$



king against king: a dead position

Lemma 3.7

$$C + T \leq 30.$$

Proof

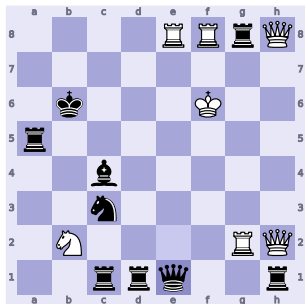
- (1) Only 30 pieces are not kings, so $C \leq 30$.
- (2) If $C \leq 29$ then $T \leq 1$ gives $C + T \leq 30$ at once.
- (3) If $C = 30$, the thirtieth capture leaves the bare kings. King against king is a dead position, so the game ends on *that* ply; no quiet ply follows and $T = 0$. □

$$\underbrace{96 + 30 - 8 + 0}_{\text{all 30 captured}} = \underbrace{96 + 29 - 8 + 1}_{\text{29, plus a closing segment}} = 118 \quad \text{neither route pays better.}$$

Theorem 3.8 — $K \leq 118$

Theorem 3.8

Every game satisfies $K = (P - O) + (C + T) \leq 88 + 30 = \mathbf{118}$. Moreover a game with $K = 118$ has $f = 8$, $P = 96$, $O = 8$, $C + T = 30$, so *all sixteen pawns move six times and promote*, and $C \geq 29$.



the real game just after its last pawn
move — no pawns

Remark 3.9 (Murphy's ceiling)

$K = 118 \Rightarrow P = 96 \Rightarrow$ both sides move pawns $\Rightarrow$ endpoints of both colours $\Rightarrow S \geq 1$, i.e.

$$L \leq 150 \cdot 118 - 1 = 17,699.$$

The last two plies are the hard part, and Part III is nothing else.

III. Switches

$K = 118$ forces $S \geq 3$

Blocks and patterns

0. Origin

I. Rules

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VI. Meaning



Definition (blocks · patterns)

Colour each critical move by whoever played it and group maximal single-colour runs into *blocks*; the list of block colours is the *pattern*. Doing the same to the endpoint sequence gives the *endpoint pattern*.

Observation

n endpoint blocks $\Rightarrow S = n - 1$ (opens Black) or $S = n$ (opens White).

So ruling out $S \leq 2$ means killing every pattern of at most three blocks.

There are exactly six patterns of at most three blocks

B		✗ only one side moves pawns	BW		✗ 14 victims never move
W		✗ only one side moves pawns	WB		✗ 14 victims never move
BWB		✗ home rank caps them at four	WBW		✗ home rank caps them at four
BWBW		✓ the only survivors: $S = 3$, $L \leq 150 \cdot 118 - 3 = 17,697$			

Kill all six and every pattern has at least four blocks, hence $S \geq 3$.

Propositions 4.1 and 4.2 — one and two blocks

Proposition 4.1 (one block)

B and W are impossible at $K = 118$.

Proof. If only one colour makes critical moves then only its eight pawns ever move, so $P \leq 48 < 96$. $\square$

Proposition 4.2 (two blocks)

BW and WB are impossible too.

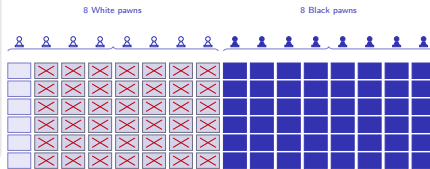
Proof. Write the pattern (X, Y) . Since Y can capture only X 's 15 capturable pieces, $C_Y \leq 15$ and hence $C_X \geq 29 - 15 = 14$. Those ≥ 14 pieces are captured *before* Y 's first critical move, so they never make a pawn move. At most 7 of them did not begin as pawns, so ≥ 7 pawns lose all six of their moves. $\square$



every critical move of X

$$C_Y \leq 15 \Rightarrow C_X \geq 29 - 15 = 14$$

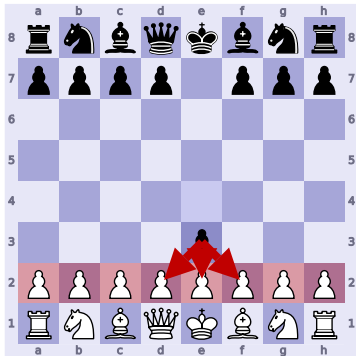
• • • • • ≥ 7 Y pawns captured, so they ne



$$P \leq 96 - 7 \cdot 6 = 54$$

< 96 $\times$

Lemma 4.3 — the home-rank lemma



1. Nf3 e6 2. Ng1 e5 3. Nf3 e4 4. Ng1 e3

On the board Black's pawn stops on e3: only White may move the pawn on e2 (violating R1), and the only way through is capturing on d2 or f2 (violating R2).

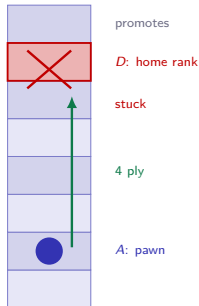
Lemma 4.3 (home rank)

Fix an attacker A and a defender D , and consider any position reachable from the initial one under

- (R1) every move by D is quiet, and
- (R2) no move by A captures a D pawn.

Then D 's pawn home rank *never changes*, no A pawn ever stands on it, and **no A pawn makes more than four pawn moves.**

Proof of Lemma 4.3



Proof

- (1) *D's home rank never changes.* Induction over moves. No move can *start* there — the square holds a *D* pawn and moving it violates (R1). None can *end* there — the square is occupied, so the move would be a capture, and *D* capturing violates (R1) while *A* taking a *D* pawn violates (R2). Castling touches only back ranks; en passant removes a pawn from rank 4 or 5.
- (2) *No A pawn stands on D's home rank.* A move ending there was ruled out in (1).
- (3) *Hence at most four moves.* The rank beyond is reachable only by a single step out of *D's* home rank or a double push from a non-home rank, both impossible. The range is four ranks and each pawn move spends one.



The induction uses only the elementary facts [H0]–[H6] about legal moves: what a move may change the occupancy of, and how a pawn advances.

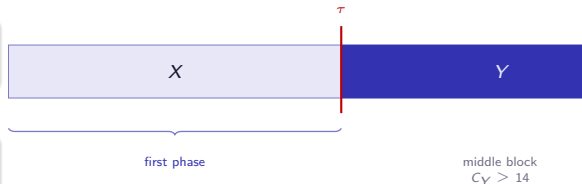
Proposition 4.4 — three blocks (X, Y, X)

Proposition 4.4

BWB and WBW are impossible at $K = 118$.

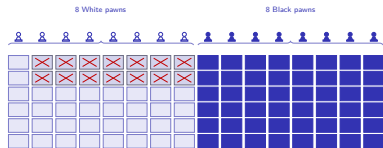
Proof

- (1) As before $C_Y \geq 14$, and all of these captures lie in the middle block. At least 7 victims are X -pieces that began as pawns.
- (2) All six pawn moves of such a piece p fall *before* Y 's first critical move τ : any X -critical move after τ belongs to the final block, which is after p was captured.
- (3) The first phase is exactly Lemma 4.3's setting — (R1) by the definition of τ , (R2) from $P = 96$. So each X pawn gets at most *four* moves there. $\square$



≥ 7 X pawns captured $\Rightarrow$ all 6×7 pawn moves fall in the first phase

but the home rank caps each pawn at 4



$$P \leq 96 - 7 \cdot 2 = 82$$

< 96 X

Theorem 4.5 — $S \geq 3$

Theorem 4.5 (four blocks)

Every game with $K = 118$ has a critical-move pattern of at least four blocks. The endpoint sequence is that sequence with at most one element appended, which cannot lower the block count, so the endpoint pattern has at least four blocks too and $S \geq 3$.



Remark 4.6. $S \geq 3$ holds *only* under $K = 118$. Repeating 1. Nf3 Nf6 2. Ng1 Ng8 four times gives a 16-ply game with $K = 1$ and $S = 0$.

Theorem 5.1 — the upper bound

Theorem 5.1 (upper bound)

No legal chess game lasts more than 17,697 plies.

Proof

$K \leq 117$ Lemma 2.4 with $S \geq 0$: $L \leq 150 \cdot 117 = 17,550 < 17,697$

$K = 118$ the largest Theorem 3.8 permits; Theorem 4.5 gives $S \geq 3$, so $L \leq 150 \cdot 118 - 3 = \mathbf{17,697}$



Corollary 5.3 (exact maximum)

The bound is attained (Labelle's generated game, Murphy's hand-built one). **The maximum length of a chess game is exactly 17,697 plies — 8,848 moves and one final White ply.**

Corollary 5.2 — the structure of a maximum-length game

Corollary 5.2

Every game of length 17,697 satisfies $K = 118$, $S = 3$, every $\delta_i = 0$, $f = 8$, $P = 96$, $O = 8$, $C + T = 30$, and both its critical-move and endpoint patterns are BWBW.

all sixteen pawns move six times, one rank each, and promote · every segment is full (150, or 149 across a change)



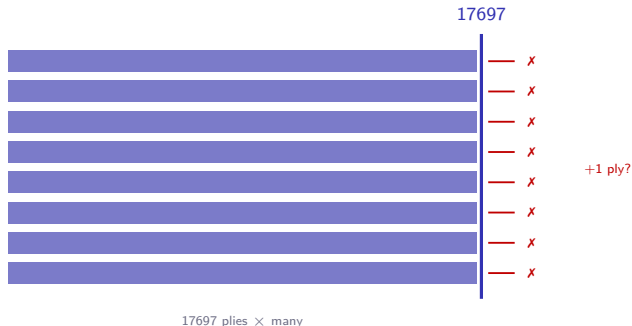
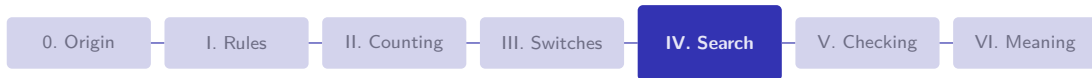
BWBW

Black makes the first critical move, and the final endpoint is White's.

IV. Search

what came before the theorem

Search 1 — brute variety



Same 118 critical moves, freshly generated quiet filler, many distinct 17,697-ply games → test every one for a single additional legal ply

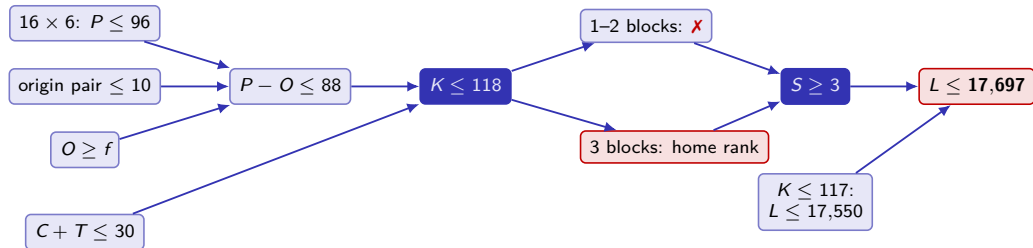
Search 2 — targeted search



Hypothesis: *a legal skeleton keeps 118 segments while switching the acting colour at most twice*

An abstract CP-SAT model over pawn-move and capture budgets, plus a planner to realise candidates as real moves — every instance infeasible

The structure of the proof

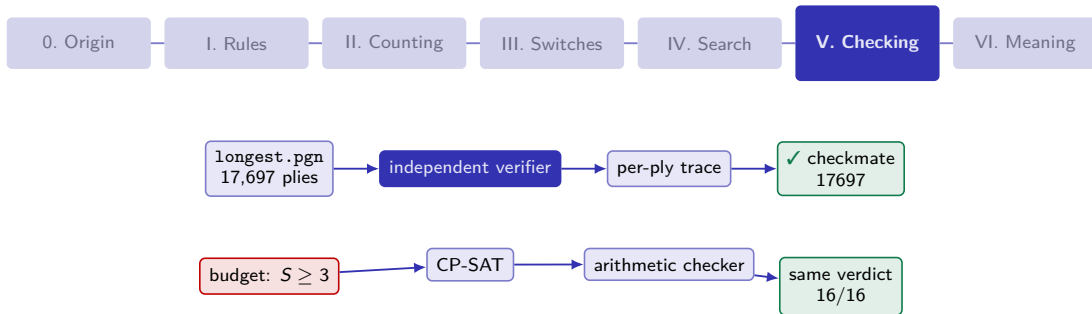


Not one line of this needs a computer.

V. Checking

computation does not replace the proof

What the computation does



Above: judge the game · Below: cross-check the hand argument by two independent methods (sound direction only — only infeasibility is used)

Two rules the verifier implements

Mate outranks the clock

The final ply is the 150th quiet ply of its segment *and* a checkmate (the precedence clause of Art. 9.6.2).

A verifier that consults the clock first accepts the same 17,697 plies but **scores the game a draw**.

Never compare whole FENs

For repetition the clock and move number play no part in identity, and en passant counts only when *legal* (Art. 9.2).

Naive keys fail both ways — a FEN over-distinguishes; placement plus side to move ignores castling rights and over-merges.

Under that identity no position of this game occurs even *three* times — fivefold repetition is never approached.

Verdict and reproduction

the game	legal, 17,697 plies, ends in checkmate
early termination	none — no position occurs even three times, no stalemate
profile	$K = 118$, $P = 96$, $C = 29$, $O = 8$, $T = 1$, $S = 3$ — exactly Corollary 5.2
decomposition	$115 \times 150 + 3 \times 149 = 17,697$, $\sum_i \delta_i = 0$
pattern	endpoint pattern BWBW
cross-check	all six patterns of ≤ 3 blocks infeasible, both methods agree

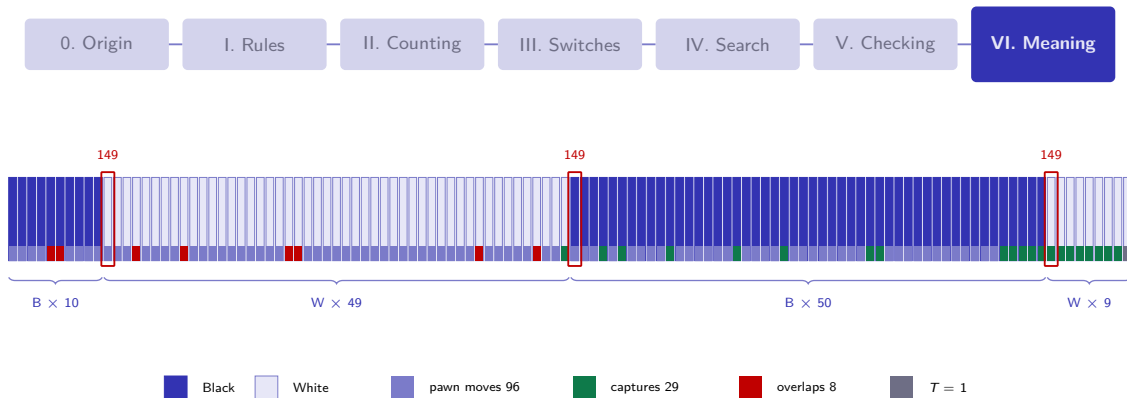
A second game is verified the same way — same 118 critical moves, fresh filler, 17,032 plies differ.

Everything re-runs from a byte-for-byte reproducible certificate: `make verify-full`

VI. Meaning

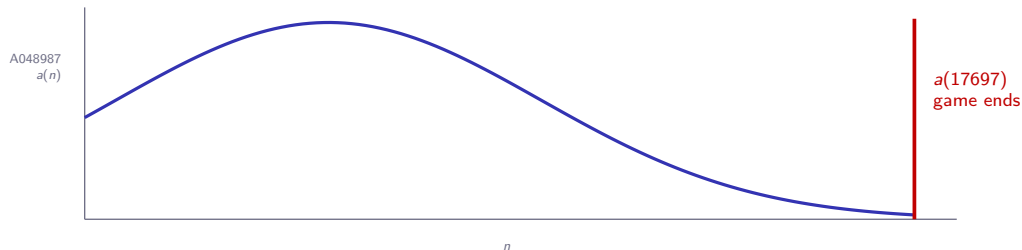
one game, and a sequence that stops

The reference game's segment structure



one bar per segment · colour is the endpoint's colour · the three 149s sit exactly on the block boundaries

Corollary — A048987 has finite support



(OEIS A048987)

The sequence counting chess games that end at ply n has **finite support**, with last nonzero term $a(17697)$.

The curve is schematic — the true $a(n)$ runs to 10^{29241} and beyond (Labelle).

Open directions

Parameter

The maximum as a function of the “ n -move rule”. The “three” in this argument uses the fact that 75 is odd.

Abstraction

Games with a bounded stock of counter-resetting events — does the same accounting hold away from chess?

Formalisation

The counting proof in a proof assistant, and re-verification under an independently implemented move generator.

The upper bound is a hand-checkable combinatorial theorem.

The lower bound is a previously known explicit construction.

Computation verifies the construction and cross-checks the proof; it does not replace it.

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github.com/junyeobe0315/long_chess

Thanks. Tom Murphy VII and François Labelle, with the forum contributors Deedlit, Tatzelwurm and Falkenbach, for the construction and the question; Alexis Langlois-Rémillard for comments on the draft; and Ing Chess, whose video started all of this.



repository



Labelle



Ing Chess



Murphy VII, SIGBOVIK 2020